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भारतीय मानक

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अनुभाग 4 चक्रण निष्कर्षक

Indian Standard

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 4 Spin Extractors

UDC 648.326.4 : 64.06-83 : 614.826

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NEW DELHI 110002

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Price Group 4

Electrical Appliances Sectional Committee, ETD 32

FOREWORD

This Indian Standard was adopted by the Bureau of Indian Standards, after the draft finalized by the Electrical Appliances Sectional Committee had been approved by the Electrotechnical Division Council.

This standard covers the general and safety requirements of spin extractors. This standard does not cover the performance requirements, which will be covered under a separate composite standard at a later date. The composite standard will also cover general safety and performance requirements of domestic electric clothes washing machines; to water extracting machines and to washing and water extracting machines, either with or without heating devices, for household use.

The requirement of this standard except for endurance test was earlier covered in IS 6390 (Part 3) : 1992 'Domestic electrical clothes washing machine — Specification : Part 3 Requirements for spin extractors'. In view of the decision of the Electrotechnical Division Council, this standard has been prepared in line with corresponding IEC Publication with latest amendment covering General and Safety Requirements. With the publication of this standard IS 6390 (Part 3) : 1992 shall be treated as withdrawn. The requirement of endurance test will be covered under the composite standard.

This standard is to be read in conjunction with IS 302-1 (1979) 'Safety of household and similar electrical appliances : Part 1 General requirements (*fifth revision*)'. For the sake of convenience, the clauses of this standard correspond to those of IS 302-1 (1979), instead of reproducing full text of each clause, clauses of IS 302-1 (1979) which are applicable (which means that relevant provisions of the clause apply) or not applicable and the sub-clauses or portions thereof which are not applicable are indicated as under:

- a) In case of a clause where it is applicable or not applicable, the wording used is 'This clause of IS 302-1 (1979) is applicable/not applicable'.
- b) In case of sub-clause or part thereof 'Not applicable'.

Wherever a sub-clause of IS 302-1 (1979) is to be replaced by a new text, it has been indicated as under:

Replacement — followed by the new text.

Any addition to the existing provisions of a sub-clause of IS 302-1 (1979) has been indicated as under:

'Addition — followed by the text of the additional matter'.

Clauses/tables which are additional to those of IS 302-1 (1979) are numbered starting from 101 and additional sub-clauses are numbered with the main clause number followed by 101, 102, etc, for example, 7.101.

Should however any deviation exist between IS 302-1 (1979) and this standard, the provisions of the latter shall apply.

While preparing this standard, assistance has been derived from the following publications issued by the International Electrotechnical Commission:

IEC Pub 335-1 (1976) Safety of household and similar electrical appliances : Part 1 General requirements

IEC Pub 335-2-4 (1984) Safety of household and similar electrical appliances : Particular requirements for spin extractors.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

The Indian Standard which is a necessary adjunct to this standard is IS 302-1 (1979) 'Safety of household and similar electrical appliances: Part 1 General requirements (*fifth revision*)'.

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Indian Standard

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 4 Spin Extractors

1 SCOPE

This clause of IS 302-1 (1979) is applicable except as follows:

1.1 Replacement

This standard applies to electric spin extractors with a load capacity not exceeding 5 kg of dry textile material and a container peripheral speed not exceeding 50 m/s, their rated voltage not exceeding 250 V, ac single phase or dc.

Spin extractors incorporated in electric washing machines are within the scope of this standard, irrespective of their load capacity.

Spin extractors not intended for normal household use, but which nevertheless may be a source of danger to the public, such as spin extractors for communal use in blocks of flats or in laundrettes, are also within the scope of this standard.

This standard does not apply to spin extractors for commercial laundries.

For spin extractors intended to be used in vehicles or on board ships or aircraft, additional requirements may be necessary.

2 TERMINOLOGY

This clause of IS 302-1 (1979) is applicable except as follows:

2.30 Replacement

Normal load denotes the load obtained when the spin extractor is operated with the container filled with a quantity of textile material having a mass in the dry condition equal to the maximum mass specified by the manufacturer.

The textile material is in the form of pre-washed double-hemmed cotton sheets having dimension of approximately 70 cm × 70 cm and a mass between 140 g/m² and 175 g/m² in the dry condition.

After weighing, the textile material is saturated with cold water before filling the container, in which it is evenly distributed.

As an alternative to the textile material specified, pieces of cloth having an area between

4 800 cm² and 5 000 cm² with one side of at least 55 cm may be used for the tests.

Additional Sub-clause

2.101 Spin Extractor — A water extracting machine in which water is removed from textile material by centrifugal force.

3 GENERAL REQUIREMENTS

This clause of IS 302-1 (1979) is applicable.

4 GENERAL NOTES ON TESTS

This clause of IS 302-1 (1979) is applicable except as follows:

Additional Sub-clause

4.101 Spin extractors are tested as portable appliances, unless they are designed to be fixed to a support or are incorporated in another appliance.

5 RATING

This clause of IS 302-1 (1979) is applicable.

6 CLASSIFICATION

This clause of IS 302-1 (1979) is applicable.

7 MARKING

This clause of IS 302-1 (1979) is applicable except as follows:

7.1 Addition

In addition, spin extractors shall be marked with the maximum mass in kilogrammes, of textile material in the dry condition, for which the spin extractor is designed, unless this is indicated in the instruction sheet.

Addition

Spin extractors must be marked with the symbol for splash-proof construction.

7.10 Addition

If the 'off' position is indicated only by a word, this word shall be 'off'.

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Additional Sub-clause

7.101 The spin extractors may also be marked with the standard mark.

8 PROTECTION AGAINST ELECTRIC SHOCK

This clause of IS 302-1 (1979) is applicable.

9 STARTING OF MOTOR-OPERATED APPLIANCES

This clause of IS 302-1 (1979) is applicable except as follows:

9.1 Addition

The test is made with the container filled with textile material saturated with water, as defined for normal load. The textile material is re-saturated with water after each operation.

10 INPUT AND CURRENT

This clause of IS 302-1 (1979) is applicable except as follows:

10.1 Addition

A representative period is a water extrusion period as specified in sub-clause **11.7** starting from cold.

11 HEATING

This clause of IS 302-1 (1979) is applicable except as follows:

11.7 Replacement

Spin extractors are operated for five periods of water extraction, the successive periods being separated by a rest period. Each rest period, which includes the braking time, has a duration of 1 minute for each kilogramme of dry textile material or 4 minutes whichever is the longer. During the rest periods, the textile material is re-saturated with cold water.

For spin extractors without a programme controller or timer, each water extraction period has a duration of:

- 15 minutes for spin extractors designed for continuous-flow rinsing; and
- 5 minutes for other spin extractors.

Unless a longer period is marked on the spin extractor or indicated in the instruction sheet, in which case the longer period applies.

For other spin extractors, the water extraction period is equal to the maximum period allowed by the programme controller or timer.

If the spin extractor is designed for continuous-flow rinsing, water is allowed to flow continuously through the textile material during the water extraction periods, in accordance with the manufacturer's instructions.

12 OPERATION UNDER OVERLOAD CONDITIONS OF APPLIANCES WITH HEATING ELEMENTS

This clause of IS 302-1 (1979) is not applicable.

13 ELECTRICAL INSULATION AND LEAKAGE CURRENT AT OPERATING TEMPERATURE

This clause of IS 302-1 (1979) is applicable.

14 RADIO AND TELEVISION INTERFERENCE SUPPRESSION

This clause of IS 302-1 (1979) is applicable.

15 MOISTURE RESISTANCE

This clause of IS 302-1 (1979) is applicable except as follows:

15.3 Replacement

Spin extractors shall be so constructed that spillage of liquid in normal use does not affect their electrical insulation.

Compliance is checked by the following tests:

Spin extractors with Type X attachment are fitted with the lightest permissible type of flexible cable or cord of the smallest cross-sectional area specified in sub-clause **26.2** and other spin extractors are tested as delivered.

The inlet to the discharge pump or to the gravity drain is blocked. The container is filled with saturated textile material as specified for the normal load, the mass of the cold water added being three times the mass of the dry textile material. Any water remaining after the saturation process is poured into the spin extractors which is then operated at rated voltage for 1 minute.

In addition, for spin extractors designed for continuous-flow rinsing and having a vertical aid a quantity of 10 l of cold water is poured into the container, which is completely filled with saturated textile material, over a period of 20 s, once while the spin extractor is operating and once while the container is at rest.

Immediately after these treatments, the spin extractor shall withstand an electric strength test as specified in sub-clause **16.4** and inspection shall show that any water which may have entered the spin extractor does not impair compliance with this standard. In particular, there

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shall be no trace of water on insulation which could result in a reduction of creepage distances and clearances below the values specified in 29.1.

The spin extractor is allowed to stand in normal test-room atmosphere for 24 h before being subjected to the test of 15.4.

16 INSULATION RESISTANCE AND ELECTRIC STRENGTH (AFTER HUMIDITY TREATMENT)

This clause of IS 302-1 (1979) is applicable.

17 OVERLOAD PROTECTION

This clause of IS 302-1 (1979) is applicable.

18 ENDURANCE (under consideration)

19 ABNORMAL OPERATION

This clause of IS 302-1 (1979) is applicable except as follows:

19.6 Addition

Spin extractors are not considered as appliances having moving parts liable to be jammed.

19.8 Not applicable

20 STABILITY AND MECHANICAL HAZARDS

This clause of IS 302-1 (1979) is applicable except as follows:

20.1 Addition

The container of the spin extractor is filled with textile material saturated with water, as defined for normal load.

Additional Sub-clauses

20.101 Spin extractors shall not be adversely affected by an unbalanced load.

Compliance is checked by the following test:

The spin extractor is placed on a horizontal support and the textile material is replaced by an object having a mass of 0.2 kg or equal to 10% of the maximum mass of textile material specified by the manufacturer, whichever is the greater. The object is fixed to a midpoint on the periphery inside the container.

The spin extractor is then operated at rated voltage or at the upper limit of the rated voltage range, for 5 min.

The test is made four times the object being moved each time through an angle of 90° around the periphery of the container.

During the tests, the spin extractor shall not overturn and the container shall not hit any part of the mechanism.

After the tests, the spin extractor shall be fit for further use.

The external enclosure is not regarded as part of the mechanism.

The container need not reach full speed during these tests.

20.102 The lid or door shall be interlocked in such a way that the appliance can only be operated when the lid or door is closed.

Compliance is checked by inspection and by manual test.

20.103 For spin extractors, whether they are separate or incorporated in a washing machine provided with a separate container for water extraction, having a motive energy not exceeding 1 500 J and a maximum peripheral speed of the container not exceeding 40 m/s, moving parts shall not be accessible when the appliance is connected to the supply or when the container rotates at a speed exceeding 60 revolutions per minute.

Compliance : is checked by inspection, by measurement and by the following test:

When the lid or door is opened by 4 mm or more, but by not more than 10 mm, it shall not be possible to touch parts rotating at a speed exceeding 60 revolutions per minute with a test pin as shown in Fig. 2 of IS 302-1 (1979), but having a length of 50 + 0.2 mm instead of 15 mm.

When the lid or door is opened by more than 10 mm, but by not more than 12 mm, it shall not be possible to touch parts rotating at a speed exceeding 60 revolutions per minute with a test rod 3 mm in diameter and 120 mm long.

In addition, the standard test finger shown in Fig. 1 of IS 302-1 (1979) is applied and shall not come within a distance of 20 mm from parts rotating at a speed exceeding 60 revolutions per minute.

When the lid or door is opened by more than 12 mm, the appliance shall be disconnected from the supply and the container speed shall not exceed 60 revolutions per minute.

If the container is not cylindrical, the peripheral speed to be considered is the mean peripheral speed.

For calculating the motive energy, the following formula is used:

$$E = 0.25 Mv^2$$

where

E is the motive energy, in joules;

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M is the mass of textile saturated with cold water and the weight of the drum material marked on the appliance is indicated in the instruction sheet, in kilogrammes; and

v is the maximum peripheral speed of the container, in metres per second.

NOTE — Manufacturer shall declare the mass of the drum on demand to the testing authority.

20.104 For spin extractors, whether they are separate or incorporated in a washing machine provided with a separate container for water extraction having a motive energy exceeding 1 500 J or a maximum peripheral speed of the container exceeding 40 m/s, it shall not be possible to open the lids or doors while the container is in motion.

Compliance is checked by inspection by measurement and by manual test.

For lids or doors, a pull not exceeding 50 N is applied in an attempt to open them as in normal use.

20.105 For spin extractors incorporated in appliances having a single container for washing and water extraction the peripheral speed of which does not exceed 40 m/s, the appliance shall be provided with an automatic means for switching off the motor, or for reducing the container speed to normal washing speed or less within 7 s after the lid or door is opened by 50 mm.

For spin extractors incorporated in appliances having a single container for washing and water extraction the peripheral speed of which exceeds 40 m/s, the appliance shall be provided with a means for preventing the opening of the lid or door while the container is in motion at a speed exceeding 60 revolutions per minute.

Compliance is checked by inspection, by measurement and by manual test. For lids and doors, a pull not exceeding 50 N is applied in an attempt to open them as in normal use.

20.106 Any safety device fitted in the upper part of a spin extractor with a vertical axis shall be so positioned or protected that the device is not likely to be damaged by textile material, which may escape from the container in normal use.

Compliance is checked by inspection.

21 MECHANICAL STRENGTH

This clause of IS 302-1 (1979) is applicable as follows:

Additional Sub-clause

21.101 Lids of spin extractors which are loaded from the top shall have adequate mechanical strength.

Compliance is checked by the following test:

A cylindrical weight having a mass of 20 kg, with a rubber hemisphere fixed to its base, is allowed to fall, three times in succession, from a height of 10 cm on to the centre of the lid in the closed position. The hemisphere has a diameter of 70 mm and a hardness between 40 IRHD and 50 IRHD.

After the test, the lid shall not be damaged to such an extent that moving parts become accessible.

21.102 Lids and doors and their hinges shall have adequate resistance to distortion.

Compliance is checked by the following test:

Lids and doors are placed in the open position and a force of 50 N is applied in the most unfavourable direction and position.

This test is made three times in succession.

After the tests, the hinges shall not have worked loose and the appliance shall not be damaged or deformed to such an extent that it fails to meet the requirements of sub-clauses 20.102 to 20.105 as appropriate.

22 CONSTRUCTION

This clause of IS 302-1 (1979) is applicable except as follows:

22.1 Replacement

Spin extractors shall be of Class I, Class II or Class III.

Compliance is checked by inspection and by the relevant tests.

22.2 Replacement

Spin extractors shall be splash-proof.

Compliance is checked by inspection and by the relevant tests.

22.3 Addition

If the instructions for installation indicate that the maximum permissible inclination of the supporting surface is 2°, the maximum angle of inclination is reduced to 2°.

23 INTERNAL WIRING

This clause of IS 302-1 (1979) is applicable.

24 COMPONENTS

This clause of IS 302-1 (1979) is applicable except as follows:

24.1 Addition

Switches incorporated in spin extractors are not required to be switches for frequent operation.

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24.3 Addition

Switches of micro-gap construction used in connection with lid or door interlocks are not regarded as intended to disconnect stationary appliances from the supply.

25 SUPPLY CONNECTION AND EXTERNAL FLEXIBLE CABLES AND CORDS

This clause of IS 302-1 (1979) is applicable.

25.6 Addition

The length of the supply cord measured between the entry point into the appliance and the entry point into the plug and cord guard included, shall not be less than 2.0 m if supplied with a plug, or shall be not less than 2.05 m if supplied without a plug.

26 TERMINALS FOR EXTERNAL CONDUCTORS

This clause of IS 302-1 (1979) is applicable.

27 PROVISION FOR EARTHING

This clause of IS 302-1 (1979) is applicable.

28 SCREWS AND CONNECTIONS

This clause of IS 302-1 (1979) is applicable.

29 CREEPAGE DISTANCES, CLEARANCES AND DISTANCES THROUGH INSULATION

This clause of IS 302-1 (1979) is applicable.

30 RESISTANCE TO HEAT, FIRE AND TRACKING

This clause of IS 302-1 (1979) is applicable.

31 RESISTANCE TO RUSTING

This clause of IS 302-1 (1979) is applicable.

32 RADIATION HAZARDS

This clause of IS 302-1 (1979) is not applicable.

101 TESTS

101.1 Type Tests

The tests specified in Table 101 shall constitute the type tests and shall be carried out on two samples of spin extractors of the same type and rating selected preferably at random from a regular production lot. Before commencement of the tests, the spin extractors shall be visually examined and inspected for obvious visual defects in respect of components, parts and their assembly, construction, mechanical hazards, marking, provision of suitable terminals for supply connections, earthing and the effectiveness of screws and connections. The external surface finish shall be even and free from finishing defects.

101.1.1 Criteria of Acceptance

Both samples shall successfully pass all the type tests for proving conformity with the requirements of the standard. If any of the samples fails in any of the type tests, the testing authority at its discretion, may call for fresh samples not exceeding twice the original number and subject them again to all tests or to the test(s) in which failure(s) had occurred. No failure should be permitted in the repeat test(s).

Table 101 Schedule of Type Tests

Sl No.	Tests	Clause Reference
i)	Protection against electric shock	8
ii)	Starting of motor operated appliances	9
iii)	Input	10
iv)	Temperature rise	11
v)	Operation under overload conditions	12
vi)	Electrical insulation and leakage current at operating temperature	13
vii)	Radio and television interference suppression	14
viii)	Moisture resistance	15
ix)	Insulation resistance and electric strength (after humidity treatment)	16
x)	Overload protection	17
xi)	Endurance (under consideration)	18
xii)	Abnormal operation	19
xiii)	Stability and mechanical hazards	20
xiv)	Mechanical strength	21
xv)	Construction	22
xvi)	Internal wiring	23
xvii)	Components	24
xviii)	Cord grip and cord guard	25
xix)	Terminals for external conductors	26
xx)	Provision for earthing	27
xxi)	Screws and connections	28
xxii)	Creepage distance and clearances	29
xxiii)	Resistance to heat fire and tracking	30
xxiv)	Resistance to rusting	31

101.2 Acceptance Tests

The following shall constitute the acceptance test:

Test	Clause Reference
a) Protection against electric shock	8
b) Input	10

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<i>Test</i>	<i>Clause Reference</i>	101.2.1 A recommended sampling procedure for acceptance tests is given in Annex B of IS 302-1 (1979).								
c) Temperature rise	11									
d) Electrical insulation and leakage current at operating temperature	13	101.3 Routine Test								
e) Moisture resistance	15	The following shall constitute the routine tests:								
f) Insulation resistance and electric strength (after humidity treatment)	16	<table><tr><th><i>Test</i></th><th><i>Clause Reference</i></th></tr><tr><td>a) Protection against electric shock</td><td>8</td></tr><tr><td>b) High voltage</td><td>13.3.2 of IS 302-1(1979)</td></tr><tr><td>c) Provision for earthing</td><td>27</td></tr></table>	<i>Test</i>	<i>Clause Reference</i>	a) Protection against electric shock	8	b) High voltage	13.3.2 of IS 302-1(1979)	c) Provision for earthing	27
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b) High voltage	13.3.2 of IS 302-1(1979)									
c) Provision for earthing	27									
g) Provisions for earthing	27									

NOTE — For the purpose of acceptance tests, the humidity treatment shall be done for 24 hours while conducting the test for moisture resistance (15).

NOTE — For the purpose of acceptance tests, the humidity treatment shall be done for 24 hours while conducting the test for moisture resistance (15).

**ANNEX A
TABLE OF TYPE TEST**

This Appendix of IS 302-1 (1979) is not applicable.

**ANNEX B
SAMPLING PROCEDURE FOR ACCEPTANCE TEST**

This Appendix of IS 302-1 (1979) is applicable.

**ANNEX C
ELECTRONIC CIRCUITS**

This Appendix of IS 302-1 (1979) is applicable if electronic circuits are used.

**ANNEX D
MEASUREMENTS OF TEMPERATURE WITH THERMOMETER**

This Appendix of IS 302-1 (1979) is applicable.

**ANNEX E
ALTERNATIVE REQUIREMENTS FOR PROTECTED MOTOR UNITS**

This Appendix of IS 302-1 (1979) is applicable,

**ANNEX F
IMPACT TEST APPARATUS**

This Appendix of IS 302-1 (1979) is applicable.

**ANNEX G
THERMAL CONTROLS AND OVERLOAD RELEASES**

This Appendix of IS 302-1 (1979) is applicable.

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ANNEX H

MEASUREMENT OF CREEPAGE DISTANCES AND CLEARANCES

This Appendix of IS 302-1 (1979) is applicable.

ANNEX J

TESTS FOR FIRE-RESISTANCE PROPERTIES

This Appendix of IS 302-1 (1979) is applicable.

ANNEX K

**BNF JET-TEST FOR DETERMINATION OF THICKNESS OF COPPER AND NICKLE
PLATING**

This Appendix of IS 302-1 (1979) is not applicable.

ANNEX L

**APPROXIMATE MEASUREMENT OF THICKNESS OF CHROMIUM ON NICKLE,
STEEL AND COPPER**

This Appendix of IS 302-1 (1979) is not applicable.

Standard Mark

The use of the Standard Mark is governed by the provisions of the *Bureau of Indian Standards Act, 1986* and the Rules and Regulations made thereunder. The Standard Mark on products covered by an Indian Standard conveys the assurance that they have been produced to comply with the requirements of that standard under a well defined system of inspection, testing and quality control which is devised and supervised by BIS and operated by the producer. Standard marked products are also continuously checked by BIS for conformity to that standard as a further safeguard. Details of conditions under which a licence for the use of the Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

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Review of Indian Standards

Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the latest issue of 'BIS Handbook' and 'Standards Monthly Additions'. Comments on this Indian Standard may be sent to BIS giving the following reference:

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